

CELL BIOLOGY – ANIMAL CELL STRUCTURE

KEY VOCABULARY

Nucleus: controls the cell and contains the genetic material (DNA).

Cytoplasm: jelly-like fluid where most reactions happen.

Mitochondria: site of aerobic respiration; releases energy.

Ribosomes: where protein synthesis takes place.

COMMON MISCONCEPTIONS

~~Animal cells have a cell wall.~~

✓ Only plant, fungal and bacterial cells have walls.

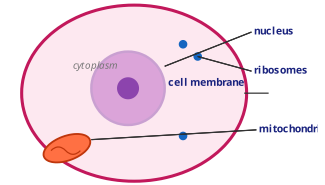
~~The nucleus makes energy.~~

✓ Mitochondria release energy in respiration.

~~All cells are the same size and shape.~~

✓ Cells are specialised for their function.

A TYPICAL ANIMAL CELL



The **cell membrane** controls what enters and leaves the cell.

1 LABEL THE CELL

Use the word bank to name the four labelled parts of the animal cell above. [4 marks]

nucleus

cell membrane

mitochondria

ribosomes

2 STATE THE FUNCTION

State the **function** of the nucleus and of the mitochondria. [2 marks]

3 ANIMAL VS PLANT

Name **two** structures found in a plant cell but not in an animal cell. [2 marks]

4 EUKARYOTIC CELLS

Explain why an animal cell is described as eukaryotic. [2 marks]

5 MAGNIFICATION

An image is 30 mm wide and the real cell is 0.06 mm wide. **Calculate** the magnification. [2 marks]

Answer: _____ ×

6 SPECIALISED CELLS

A sperm cell has many mitochondria. **Suggest** why. [2 marks]